ADA-Marker Answer Sheet

Anonymized local demo

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Answer

```
5. palindrome (S, n)
    int left=0, right=n-1, max=-∞, score=0
for(int i=0, i<n; i++)
    left=i
    right=n-1, score=right-left
    while(left<right)
        if (S[left]==S[right])
                left ++
                right--
        else
                right--
                score--
    max=max(max, score)
return max
```

First Set the score = possible max length So we use a nested loop to traverse the string, we start from the first bit, if match then compare next if not then make the right backward, and score -- (like skip or delete a letter until match. so obviously the outer loop is $O(n)$, the while will done this $O(n)$ times from left to right.

So $O(n) \cdot O(n) = \underline{O(n^2)}$ #